

## RICH EARTH

If the mineral and organic ingredients are in the right proportions, they combine to form a fertile, crumbly soil called loam. Such soils are ideal for plant growth because they contain plenty of plant nutrients and rarely become waterlogged.



### ▲ PLOWLAND

*Plowing the land folds plant remains back into the soil, where they decay and add to its fertility.*



## HIDDEN LIFE

Healthy soil is alive with microbes, which support worms and creatures such as burrowing centipedes. These help convert the remains of dead plants and animals into plant nutrients. Earthworms also mix up the soil and allow air to penetrate, which encourages the growth of microbes.



## VITAL MINERALS

Plants need nutrients. They get some from decayed organic matter, but others are minerals released from rocks by weathering.

Crumbling volcanic rocks create rich soils, which is why people risk their lives to farm on the flanks of active volcanoes. Here, a vineyard grows on the volcanic slopes of Mount Pico, in the Azores.



## RAIN AND SAND

High rainfall or rapid drainage through sand or gravel can make soils less fertile. The water washes soluble plant nutrients and iron out of the topsoil and deposits them lower down. The soil develops distinct layers, with an acidic, infertile layer where the iron has been washed out. Only specialized plants such as heathers thrive in these acid soils.

